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Joint Radar Emitter and Mode Clustering Using a Transformer Encoder Architecture

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Abstract

This thesis investigates joint clustering of radar emitters and their operating modes from intercepted pulse descriptor word (PDW) streams using a transformer encoder architecture.

Keywords: transformer, deep clustering, electronic intelligence, radar emitter identification, mode recognition, self-supervised learning.

Sammanfattning

Detta examensarbete undersöker gemensam klustering av radarsändare och deras operativa moder från avlyssnade pulsbeskrivande ord (PDW) med hjälp av en transformerbaserad encoderarkitektur.

Nyckelord: transformer, djup klustering, signalspaning, radarsändaridentifiering, modigenkänning, självövervakad inlärning.

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Markus Swegmark

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List of Acronyms

ELINT electronic intelligence

ESM electronic support measures

EW electronic warfare

PDW pulse descriptor word

CHAPTER 1

Introduction

1.1 Background and Motivation

Start with story about Ukraine and Information operations (IO).

Electronic warfare (EW) is becoming increasingly important in modern warfare as technological advancements continues in radar technology, and the advancement of computer algorithms, such as machine learning, which can be superior to humans at some delimited tasks.

One of the main objectives of electronic warfare is to control the electromagnetic spectrum. Identifying radars in your surroundings is a crucial part of this. In the subset of EW called electronic support measure (ESM), where one passively listens to the electromagnetic environment to infer and deduce useful information about it, especially information about other agents in it, including mapping which radars are present and information about them.

In modern electronic warfare, so called agile radars, which can be anything from missiles to airplanes to stationary radars, are becoming more and more sophisticated. When emitting a signal for detection, they are intentionally varying the characteristics of the signal, like frequency, pulse width, amplitude, pulse repetition interval to hide their presence in an increasingly more unpredictable way and in expanding ranges, enabled by modern hardware and evolving algorithms (including machine learning).

This leads to a “cat and mouse”-like situation, where the agile radar agent wants to either scan an area, or lock into a target, sometimes being the entity with the ESM, or something else, without being detected themselves, or in other ways try and confuse the ESM.

Apart from having different algorithms for varying the pulse signal, the agile radars also send out different signals depending on the objective, for example if it’s scanning it might have a radar rotating around an axis, leading to an oscillating received signal, while if the radar wants to track a single object it can send a more direct signals, leading to a constant signal in the naïve case. Usually, the algorithms for varying the signal characteristic are pretty simple, maybe it changes frequency in a set sequence or random etc, and keeps it like this. However, this can will make it hard to identify and also more advanced algorithms of course exists. This algorithm configuration for varying the signal characteristics combined is what we refer to a mode of the radar (this includes if it’s searching or locking on). If we know the mode of a radar, we basically know how the signal will look like (plus minus some noise and interference etc), and also it tells us about the agent sending out the signal if it’s searching or locking in, and also can

be used to distinguish between emitters and tell us of how advanced the radar is and perhaps even the goal of the agent. Because of this, it would be ideal to identify not only what signals comes from what radar, but also what modes they are using. Note that two radars can have same modes, but also change the modes. We might need to make some limitations and assume that one radar stays in the same mode. Our input data also give us angle of incoming signal which should enable us to filter or give support for classifying different radars from positioning.

In the field of ES we usually call it radar classification (RC) if we want to classify what type of emitter the pulses are coming from. E.i SA-6 radar, drone XXY, etc. While if we want to say exactly what instance it is, that is also distinguish the number of emitters from each emitter as well, e.i SA-6 radar 3, drone XXY 1 and 3, we call that specific radar identification (SRI). So you can say that RC is a superset of SR, and since it provided more information it is more advantageous - however the marginal strategic gain from information of SRI compared to RC can be discussed while the marginal cost might be discussed. The marginal gains is we know the number of emitters exactly, and the ID hopefully, however this problem is sometimes harder and might have lower accuracy compared to pure RC, thus decreasing the Marginal gains. The marginal cost is however that our model might get more complicated, and depending on our model, it could also decrease the accuracy of the classification problem - we will not allow this to happen.

1.2 Problem Statement

Modern radar systems are evolving rapidly. They can change their transmission behavior on the fly (often called “agile” radars), which makes it increasingly difficult for traditional, rule-based methods to classify and identify them reliably. In Electronic Warfare (EW), Electronic Support Measures (ESM) systems passively listen to the electromagnetic environment and aim to detect, characterize, and identify radar emitters in order to enhance situational awareness and protection. Radars emit short bursts of energy known as pulses. Different radars tend to produce characteristic patterns across pulse features such as carrier frequency, pulse width, time of arrival, and angle of arrival. By analyzing these patterns, it is possible to infer how many emitters are present and which emitters they are. However, overlapping signals, agile behavior, noise, and dynamic scenarios make this a complex pattern recognition problem.

This thesis addresses the end-to-end ML classification problem of determining, from recorded radar pulses, both the number of radar emitters present in a flight simulation and their identities. The work will use labeled, simulator-generated ESM data provided by Saab, in which each pulse is described by features such as frequency, pulse width, time of arrival (ToA), and angle of arrival (AoA). The problem is formulated as a supervised learning task, with the goal of training models that map pulse observations to emitter identities. For this thesis you will both design an AI model but also analyze the

simulated data. To understand how good the model is it is also required to understand the difficulty of the data it is evaluated on. You will create metrics for measuring dataset complexity and then compare how your model performs on data of varying complexity according to your metrics.

1.3 Research Questions

The thesis addresses the following research questions:

RQ1 Placeholder research question.

RQ2 Placeholder research question.

1.4 Objectives and Scope

Placeholder paragraph.

1.5 Contributions

The main contributions of this thesis are:

- Placeholder contribution.
- Placeholder contribution.

1.6 Ethical and Societal Considerations

Placeholder paragraph.

1.7 Thesis Outline

The remainder of this thesis is organized as follows. Chapter 2 introduces the necessary background on radar signal processing and deep learning. Chapter 3 surveys related work. Chapter 4 describes the proposed method. Chapter 5 presents experiments and results. Chapter 6 discusses the findings. Chapter 7 concludes and outlines future work.

CHAPTER 2

Background

2.1 Radar Systems and Passive Reception

Radio is the art of communicating using electromagnetic waves, traditionally done, but not limited to, radio waves. Discovered in theory in the electromagnetic revolution in the 1800s by the great Maxwell, and made reality in the end of that century by Guglielmo Marconi who in 1891 transmitted the first morse code over a kilometer using radio waves from his device. xt

Radar, which is an abbreviation for radio detection and range, is a form of echolocation, using radio waves. It was discovered in the early 1900s, and its development was greatly boosted due to its utility detecting planes and warships in world war 2.

2.2 Radar Emitters and Operating Modes

Placeholder paragraph.

2.3 Pulse Descriptor Words

Placeholder paragraph.

2.3.1 EW Signal Processing Pipeline

In EW and ESM contexts, the path from the electromagnetic environment to downstream reasoning is often described as a *signal processing pipeline*. At a high level, wideband receivers capture emissions, detectors mark pulse events, and estimators extract per-pulse parameters such as carrier frequency, pulse width, and time of arrival. Deinterleaving then groups pulses into coherent trains and emits PDWs (or equivalent feature vectors) as the interface to higher-level ELINT tasks, including emitter and mode analysis. Placeholder paragraph.

2.4 Machine Learning Fundamentals

Placeholder paragraph.

2.5 Clustering Methods

Placeholder paragraph.

2.6 Deep Representation Learning

Placeholder paragraph.

2.7 The Transformer Encoder

Placeholder paragraph.

CHAPTER 3

Related Work

3.1 Classical Radar Emitter Identification

Placeholder paragraph.

3.2 Deep Learning for Radar Signal Analysis

Placeholder paragraph.

3.3 Deep Clustering

Placeholder paragraph.

3.4 Transformers for Time Series and Sets

Placeholder paragraph.

3.5 Joint and Multi-Task Clustering

Placeholder paragraph.

CHAPTER 4

Method

4.1 Problem Formulation

Placeholder paragraph.

4.2 Data Representation and Preprocessing

Placeholder paragraph.

4.3 Transformer Encoder Architecture

Placeholder paragraph.

4.4 Loss Function and Clustering Objective

Placeholder paragraph.

4.5 Training Procedure

Placeholder paragraph.

4.6 Evaluation Metrics

Placeholder paragraph.

4.7 Baseline Methods

Placeholder paragraph.

CHAPTER 5

Experiments and Results

5.1 Experimental Setup

Placeholder paragraph.

5.2 Datasets

Placeholder paragraph.

5.3 Hyperparameters and Training Details

Placeholder paragraph.

5.4 Quantitative Results

Placeholder paragraph.

5.5 Ablation Studies

Placeholder paragraph.

5.6 Qualitative Analysis

Placeholder paragraph.

CHAPTER 6

Discussion

6.1 Interpretation of Results

Placeholder paragraph.

6.2 Comparison with Related Work

Placeholder paragraph.

6.3 Limitations

Placeholder paragraph.

6.4 Implications

Placeholder paragraph.

CHAPTER 7

Conclusion

7.1 Summary

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7.2 Answers to the Research Questions

Placeholder paragraph.

7.3 Future Work

Placeholder paragraph.

APPENDIX A

Additional Results

Placeholder paragraph.

APPENDIX B

Implementation Details

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APPENDIX C

Notation Reference

Placeholder paragraph.